

BARUN SAHA

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Professional Summary

B.Tech Computer Science and Engineering student with a strong interest in software engineering and application development. Experienced in building software solutions using Java, Python, C++, web technologies, databases, and modern development tools. Passionate about designing scalable applications, solving complex problems, and continuously learning new technologies.

Education

Adamas University

B. Tech. in Computer Science and Engineering

Kolkata, India

2028

Saint Paul's School

Higher Secondary Education — Science

Jalpaiguri, India

2023

Relevant Coursework

Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering

Technical Skills

Languages: Python, C, C++, Java

ML / AI: PyTorch, Transformers, ViT, CLIP, BLIP, Qwen2-VL, LoRA/QLoRA

Computer Vision: OpenCV, Grad-CAM, SAM, YOLO

Remote Sensing: GDAL, Rasterio, GeoPandas, GeoTIFF, Optical + SAR

Web / Backend: Spring Boot, REST APIs

Databases: MySQL MongoDB, Redis

Tools & DevOps: Git, GitHub, Docker, Kaggle, Google Colab

Core Concepts: DSA, OOP, DBMS, System Design, Machine Learning

Technical Experience

IEEE CIS Kolkata Chapter | Summer Intern

Kolkata

June 2026 – August 2026

- Worked on AI/ML research involving Vision Transformers, CLIP, multimodal models, and remote-sensing imagery.
- Explored optical and SAR data, visual question answering, change detection, and model interpretability techniques.

Projects

SatQuery AI | *PyTorch, Qwen2-VL, QLoRA, Optical + SAR, VQA*

- Built a bi-temporal remote-sensing VQA system for querying satellite imagery and analyzing land-cover changes.
- Integrated optical and Sentinel-1 SAR workflows with multimodal language-model reasoning and change-detection outputs.
- https://github.com/barun20saha/sih_SatQuery

Annapurna — Smart Farming Assistant | *AI/ML, Computer Vision, Geospatial Data*

- Designed a smart agriculture assistant concept combining machine learning, geospatial information, and crop-focused insights.
- <https://github.com/barun20saha/classifier-disease>

SafeHer | *React, Backend APIs, TypeScript*

- SafeHer is a smart women's safety platform offering SOS alerts, live tracking, AI voice assistance, evidence capture, and discreet emergency protection.
- <https://github.com/SafeHer-Team/SafeHer->

MyAccounts | *Android, Java/Kotlin, Local Storage*

- Designed a personal money-management application focused on recording expenses and tracking finances.

Certificates

- NPTEL — https://drive.google.com/file/d/110gMNLAXbA6Sx0fp9NzMHVtXLDi_KR6e/view?usp=sharing
- Data Science Course — https://drive.google.com/file/d/1UsCLcmp5oO4FYjRWtDGo1Oagdh7_UHaW/view?usp=sharing

Selected Achievements

- Smart India Hackathon (SIH) — Team “Red Orbit”
- IEEE CIS Kolkata Chapter Summer Internship 2026
- Data Science / AI-ML project work
- Leetcode — 200+ solved